

OPPONG KWABENA ADUTWUM

CONTACT

✉ oppongkwabena1777@gmail.com
☎ +233 53 529 9655
📍 Kumasi, Ghana
➤ [linkedin.com/in/kwabena-opping-adutwum](#)
➤ [github.com/Oppong-pv](#)
➤ [opping-pv.github.io](#)

TECHNICAL SKILLS

Programming Languages

- Python
- JavaScript
- C++
- HTML / CSS
- SQL

Data & Analytics

- Pandas / NumPy
- SQL / PostgreSQL
- Jupyter Notebooks
- Matplotlib / Seaborn
- Data Preprocessing & EDA

AI & LLM Tools

- Scikit-learn
- Groq AI / Anthropic Claude API
- LLM Prompt Chaining / RAG
- Supervised Learning
- Neural Networks (fundamentals)
- PyTorch
- TensorFlow

Web & Backend

- React Native / Expo
- Django / FastAPI / Flask
- SQLite (offline-first)
- REST API Design

Dev Tools & Design

- Git / GitHub
- VS Code / CLion
- Figma
- Linux (Kali)
- Arduino IDE
- Expo CLI

Core Knowledge

- Machine Learning Fundamentals
- Network Security / Ethical Hacking
- Embedded Systems / IoT
- PID Control / Robotics
- Data Structures & Algorithms
- OSI & TCP/IP Networking

LANGUAGES

- English — Native
- French — Intermediate
- Twi — Native

CERTIFICATIONS

- Hashgraph Developer Course — Hedera (Jan 2026)
- Hedera Certified Foundation (Apr 2026)
- IPPF Competitor — Brewer Foundation & NYU (2024-25)
- WASSCE General Science — Prempeh College (Sep 2025)
- [Deep Learning Spec.] — DeepLearning.AI
- [ML Specialization] — Stanford / Coursera

GITHUB HIGHLIGHTS

- arethos — Mental wellness app
- [repo] — CNN crop disease classifier
- [repo] — LLM study assistant

MACHINE LEARNING ENGINEER

Applied ML · Robotics · Embedded Intelligence · Full-Stack Development

PROFESSIONAL SUMMARY

Computer Science undergraduate at Kwame Nkrumah University of Science and Technology (KNUST) with hands-on experience in embedded intelligence, autonomous systems, and applied machine learning. National Robotics Champion (Robofest Ghana 2025) and Regional Cybersecurity Champion (2024). Building intelligent systems — from PID-controlled autonomous robots to AI-powered mobile applications.

Combining strong fundamentals in computer science, mathematics, and algorithmic thinking with practical engineering experience across React Native mobile development, Django backend systems, and dual-AI architectures (Groq + Anthropic). Seeking an AI/ML internship to design, train, and deploy models that solve real-world problems at scale.

EXPERIENCE

Co-Founder & Software Developer — Arethos Tech Initiatives

Kumasi, Ghana | 2026 – Present

- Co-built Arethos, a cross-platform mental wellness app for KNUST students: AI-powered journaling, mood tracking, and reflections (React Native, Expo SDK 54, SQLite, Groq + Anthropic AI dual-fallback).
- Engineered dual-AI fallback architecture (Groq primary, Anthropic secondary) ensuring 99%+ uptime for the reflection engine with automatic failover and retry logic.
- Built custom haptic feedback system with multi-pulse waveforms, calibrated through 9 iterations on Samsung Galaxy S23. Solved expo-haptics Android limitation by using `Vibration.vibrate()` with `createWaveform`.
- Implemented PanResponder-based sleep slider with ref-pinned state to eliminate stale-closure bugs in gesture handlers.
- Designed high-fidelity Figma component design systems; coordinating sprint tasks via Git.
- Leading the ACES CodeFest 2026 UI/UX Challenge entry — redesigning the ACES website into a mobile-first experience across 9 pages in a 2-week sprint.

Machine Learning Trainee — Zap Tech Technologies

Remote | May 2025

- Completed an intensive 2-week ML engineering program covering the full model development lifecycle — from raw data collection through model evaluation and iteration.
- Preprocessed and cleaned real-world datasets using Pandas and NumPy, handling missing values, feature scaling, and data augmentation to improve model input quality.
- Trained and evaluated classification models using Scikit-learn, interpreting accuracy, precision, recall, and F1-score to compare model performance across iterations.
- Documented all experiments, hyperparameters, and results in structured notebooks — building the discipline of reproducible ML research from day one.

Online Safety Analyst & Volunteer — Solutions for Life Initiative Ghana (SFLIG)

Ghana | Oct 2022 – Present

- Joined as a volunteer in October 2022 for community support projects donating food, sanitary items, and clothing to rural communities.
- Promoted to Online Safety Analyst (August 2023), managing the organisation's social media and website with a focus on internet security.
- Runs speaking engagements during semester breaks with girls from orphanages and rural communities on online safety topics.

[AI/ML Engineering Intern] — [Company — e.g. Anthropic, Google DeepMind]

[City, Country] | [Mon YYYY – Mon YYYY]

- Trained and fine-tuned [model type] on [dataset], improving [metric] by [X]% over baseline.
- Built end-to-end ML pipeline: data ingestion → preprocessing → training → evaluation → deployment on [AWS/GCP].

EDUCATION

Kwame Nkrumah University of Science & Technology (KNUST)

B.Sc. Computer Science | Kumasi, Ghana | Expected 2029

Relevant Coursework:

- Data Structures and Algorithms — Arrays, linked lists, trees, graphs, sorting, dynamic programming, complexity analysis.
- Discrete Mathematics — Logic, set theory, combinatorics, graph theory, proof techniques.
- Probability and Statistics — Probability distributions, Bayesian inference, hypothesis testing, regression analysis.
- Networking & Internet — OSI/TCP/IP models, IPv4 addressing, subnetting, network protocols.
- Calculus I — Limits, derivatives, integrals, applications to optimisation.
- C++ / Computer Science & Mathematics Programming.
- Systems Programming — Memory management, process control, file I/O.

INTERESTS

- Mathematics
- Machine Learning & AI
- Robotics & Embedded Systems
- UI/UX & Visual Design
- Movies & Video Games
- Sports (Football)

REFERENCES

- Emmanuel Owusu Ansah
Senior Teacher, Maths — Prempeh College
+233 54 538 3894
- Kwadwo Gyan-Darkwa
Physics Teacher — Prempeh College
+233 244 940 819
- Sally Anokye-Yeboah (Mrs)
Executive Director — SFLIG
+233 559 1864 / +233 244 212 953

Self-Directed Study:

- Andrew Ng's Machine Learning Specialization (Stanford / Coursera) — supervised learning, neural networks, recommender systems.
- Deep Learning Specialization (DeepLearning.AI) — CNNs, RNNs, sequence models, optimisation.
- Competitive programming practice on LeetCode and Codeforces.

Prempeh College (Senior High School)

WASSCE — General Science | Kumasi, Ghana | Graduated September 2025

- Core subjects: Core Mathematics, Elective Mathematics, Physics, Chemistry, English Language, Integrated Science, ICT.

Extracurricular Leadership:

- Founded and led the Prempeh College Cybersecurity Club (2023–2025). Organised weekly sessions on network security, ethical hacking, and CTF challenges. Mentored 6 junior students.
- Active member of the Robotics Club (2023–2025). Won the Robofest Ghana 2025 National Championship.
- Represented Prempeh College at IPPF (2024–2025), organised by Brewer Foundation and NYU.

◆ PROJECTS

Arethos — Mental Wellness App (3-Person Team)

React Native · Expo SDK 54 · SQLite · Groq & Anthropic AI | github.com/Oppong-py/arethos

- Co-built a cross-platform mental wellness app for KNUST students with three core tabs: AI-powered Journal, Mood tracker, and History/Reflections.
- Engineered dual-AI fallback (Groq + Anthropic) ensuring 99%+ uptime. Built custom haptic feedback calibrated through 9 iterations.
- Implemented PanResponder-based sleep slider with ref-pinned state to eliminate stale-closure bugs in gesture handlers.
- Built Modal-based bottom-sheet delete system, bypassing React Native's unreliable zIndex between non-sibling elements.
- Voice recording with real-time waveform visualisation. Full dark/light theme toggle.
- SQLite for offline-first local storage — no user data ever leaves the device.
- Authored 16-chapter technical documentation covering every architectural decision.

Autonomous Line-Following Robot

LEGO MINDSTORM EV3 · PID Control Algorithm

- Designed PID control algorithm for autonomous line navigation — National Champion at Robofest Ghana 2025.
- Tuned Kp, Ki, Kd through systematic testing. Achieved sub-second course correction at high speed.
- Optimised sensor-motor feedback loops under strict competition time limits — directly applicable to reinforcement learning and robotics control.

Class Attendance Monitoring System

React Native · Django · PostgreSQL · Bluetooth

- Architected biometric attendance platform serving 50 students in prototype testing, reducing manual roll-call by ~80%.
- Backend: Django REST Framework with PostgreSQL. RESTful API for registration, attendance logging, analytics.
- Frontend: React Native mobile app for cross-platform deployment.
- Hardware: BLE communication for proximity-based check-in via fingerprint scanner.

Solar-Powered IoT Watering System

Arduino · IR Sensor · C++ · Solar

- Designed sensor-driven embedded system with proximity-triggered actuation — edge computing for IoT agriculture.
- Full circuit: solar charge controller, relay logic, IR sensor array, C++ firmware on Arduino.
- Solar power with voltage regulation for off-grid operation.

[Computer Vision — Crop Disease Detector]

[PyTorch · CNN/ResNet · FastAPI · AWS]

- Train CNN on labelled plant images for disease detection. Production web app for farmer diagnosis.

[NLP / LLM — AI Study Assistant]

[Claude/OpenAI API · LangChain · RAG · Next.js]

- RAG-powered chatbot for textbook Q&A with safety guardrails.

◆ HACKATHONS & COMPETITIONS

National Cybersecurity Challenge — Regional Champion (2024)

- Knowledge-based competition: network security, threat analysis, incident response, cryptography. Represented Prempeh College across the Ashanti Region.

Robofest Ghana — National Champion — Robotics (2025)

- Design, build, programme autonomous robots for timed tasks. Won National Championship against teams from across the country.

ACES CodeFest 2026 — UI/UX Challenge — Upcoming (July 2026)

- Competing in the ACES KNUST CodeFest UI/UX challenge — redesigning the ACES website into a mobile-first experience across 9 pages in a 2-week sprint.

International Public Policy Forum (IPPF) — Competitor (2024–2025)

- Global debate competition by Brewer Foundation & NYU. Critical analysis, evidence-based argumentation.

◆ AWARDS & HONOURS

- National Champion — Robofest Ghana 2025 (Autonomous Robotics)
- Regional Champion — National Cybersecurity Challenge (2024)
- IPPF Competitor — Brewer Foundation & NYU (2024–2025)
- Hedera Certified Foundation (Apr 2026) — among the earliest blockchain-certified students at KNUST.
- Strong academic performance in Mathematics-heavy coursework: Calculus I, Probability & Statistics, Discrete Mathematics, C++ Programming.

◆ LEADERSHIP & VOLUNTEERING

Club Leader — Prempeh College Cybersecurity Club (2023–2025)

- Founded the club. Weekly sessions on networking, ethical hacking, Kali Linux, Wireshark. Mentored 6 junior students.

Online Safety Analyst & Volunteer — SFLIG (Oct 2022–Present)

- Promoted from volunteer to analyst. Manages social media, website, and runs speaking engagements on online safety.

Participant — International Public Policy Forum (IPPF) (2024)

- Global policy debate. Research, argumentation, and collaborative problem-solving with international peers.

◆ MEMBERSHIPS & COMMUNITIES

Member — Association of Computer Engineering Students (ACES), KNUST (2023–Present)

Member — AmaliTech Coding Club (2025–Present)

Member — Microsoft User Group Ghana — KNUST Chapter (2025–Present)

Member — Claude Builder Club — KNUST (2025–Present)

Member — Prempeh College Robotics Club (2023–2025)

◆ PERSONAL DEVELOPMENT

- Portfolio website: oppong-py.github.io — built with vanilla HTML/CSS/JS, hosted on GitHub Pages.
- Technical blog posts: 'From PID Control to Reinforcement Learning' and 'Three Bugs That Taught Me More Than Any Tutorial'.
- *[Open-source contributions — add as you contribute]*

◆ ADDITIONAL INFORMATION

- Nationality: Ghanaian
- Location: Kumasi, Ghana (open to relocation for internships)